

Errors reveal the depth of human reasoning

Tamal Maharaj ^{1*} and Kenneth W. Regan ²

¹Ramakrishna Mission Vivekananda Educational and Research
Institute, Belur Math, West Bengal, India.

²Department of Computer Science and Engineering, University at
Buffalo (SUNY), Buffalo, New York, USA.

*Corresponding author(s). E-mail(s): tamal@gm.rkmvu.ac.in;
Contributing authors: regan@buffalo.edu;

Abstract

How deeply people think before choosing is the hidden quantity at the centre of bounded rationality, yet rarely measured in natural behaviour. We read it from the structure of human mistakes: across more than a million skill-graded chess decisions scored by an engine at every search depth, the moves of strong and weak players look comparably good to a shallow analysis and separate only as it deepens—what divides them is the depth at which their errors are exposed, the depth of satisficing. Under an estimator given no rating information at all, it rises with skill in slow play and is absent in the fastest games; it recovers the known depth of synthetic agents and improves move prediction only where deep reasoning has room to matter. Strong and weak players’ errors differ not in size but in how deep one must look to see them. The central claims survived a prospective replication registered before its data existed; one registered claim—that inferred depth tracks thinking time—is withdrawn here as an artefact of the model’s prior. What people get wrong, more than what they get right, reveals how they think.

Keywords: bounded rationality, satisficing, depth of reasoning, decision-making, expertise, chess

Author note.

Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is split by player throughout. T. Maharaj formerly published as T. T. Biswas.

047 1 Introduction

048
049 Seventy years of decision science rest on a quantity nobody can see. When Simon pro-
050 posed that people *satisfice*—search until an option clears an acceptability threshold,
051 then stop—he made the depth of that search the load-bearing variable of bounded
052 rationality [1, 2]. The same hidden quantity does the work in the fast-and-frugal pro-
053 gramme [3], in resource-rational analysis, which treats cognition as the optimal use
054 of limited computation [4–7], in heuristics-and-biases accounts of judgement [8, 9],
055 and in the level- k and cognitive-hierarchy models of strategic thinking [10–16]. Yet in
056 practice, how far a person looks before committing has been inferred almost entirely
057 from small staged experiments. A notable recent exception used a controlled board
058 game with behavioural modelling to show that planning depth grows with expertise
059 [17]. Whether such regularities hold in real, repeated, consequential choice—outside
060 the laboratory and at the scale of populations—has been hard even to ask, because
061 natural decisions rarely come with a ground truth for how good each option was, let
062 alone for how good it *looked* at each successive depth of consideration.

063 Chess supplies exactly that, which is why it has served for a century as the model
064 organism of expert cognition [18–22]. Millions of decisions by players of precisely known
065 skill are recorded in real competition, and every one of them can be scored by an engine
066 far stronger than any human [23, 24]—not once, but at every depth of an iterative-
067 deepening search. The engine’s opinion of a move changes as it looks deeper: a move
068 that seems strong at a glance may be refuted only far down the tree, and a move that
069 looks dull at first may reveal its point three moves later. This depth-resolved record
070 lets us put a new question to every human mistake: *at what depth does the flaw become*
071 *visible*? And therefore: how deep must the player have looked?

072 Whether expertise involves searching deeper at all is itself an old and unsettled
073 question in this model system. De Groot’s think-aloud protocols found that grand-
074 masters and strong club players searched to similar depths and considered similar
075 numbers of moves; what set the masters apart was that they looked at the right moves
076 [18]. Later studies with more players and finer measures found that depth does grow
077 with skill [25–27], though modestly and unevenly [28], and a re-examination seventy
078 years on concluded that search and pattern knowledge are complementary rather than
079 rival explanations [29]. Every one of these studies rests on a few dozen players think-
080 ing aloud over a handful of positions. Engines have since been used to score human
081 moves against a near-perfect standard [30–32], but the depth at which a human’s error
082 becomes visible—the quantity the classic debate turns on—has not been read from
083 that standard at scale.

084 We call the answer the **depth of satisficing**—the depth at which the move a
085 player chose stops looking better than the move they should have played. Our central
086 empirical claim is that this is the dimension along which skill is most legible. Judged
087 only at shallow depth, the moves of players separated by a thousand rating points look
088 comparably good; the gap opens as the engine searches deeper. What divides them
089 is carried by their errors—not by how many they make, but by how deep one must
090 look before the flaw appears. Skill is present even in moves played at a glance, since
091 stronger players recognise better options without searching; but most of what rating
092 buys is realised only through the depth to which a player effectively searches.

Online chess also supplies a natural, graded manipulation of how deeply a person can afford to look: the time control. A player with an hour for the game can follow a line many moves ahead; one with ten minutes cannot; and one playing *bullet*—a minute or two for all the moves—has time only to react. Because the same players compete across every control, the clock varies the budget for search while holding the person, and their pattern knowledge, fixed. If the depth of satisficing measures search rather than mere skill, it should track that budget: deepest in classical play, shallower in rapid and blitz, and collapsing in bullet, where even strong players must move on recognition alone with no time to verify what they see. This graded prediction across four controls is among the sharpest tests we can put to the construct, and its limiting case is a built-in control: in bullet, search is unavailable to everyone, so any residual skill difference there is pattern, not depth.

Three things distinguish this work from earlier studies of chess errors and engine depth [31, 33, 34]. First, we replace descriptive curve-fitting with a generative model: the player is a latent distribution over search depths, so the depth of satisficing is a posterior quantity with stated uncertainty rather than the crossing point of two fitted curves. Second, we do not assume that engine depth is a valid proxy for human reasoning depth; we test it—predictively, differentially, against planted ground truth, and under estimators given no rating information—and we report where it fails. Third, we subjected the construct to an adversarial replication programme: hypotheses, pipeline, and decision rules were pre-registered and then confirmed on untouched months of data, ending with a fully prospective test registered before its data existed [35]. The final, registered confirmation succeeded on all three primary hypotheses.

2 Results

Reading a decision at increasing depth.

A chess engine evaluates by *iterative deepening*: it scores every candidate move looking one ply (half-move) ahead, then two, and so on, revising each move’s evaluation at every new horizon; evaluations change with depth because deeper search uncovers consequences that shallower search cannot see—the game-tree analogue of thinking further ahead. A move’s value is therefore not a number but a *trajectory* across depth, and that trajectory is our raw material. Figure 1a shows a real decision from the data. A 1910-rated player, offered a bishop, recaptured it (fxe6): at shallow depth the capture looks close to winning, but from about eight plies the refutation comes into view and its value collapses. The quiet king move Kb8 shows the opposite signature—unimpressive at a glance, it proves best once the search runs deep. Writing $\delta_{m,d}$ for move m ’s *regret* at depth d —the win probability it concedes relative to the best option visible at that depth (Methods)—we call the signed area between a move’s regret trajectory and its full-depth endpoint the move’s *swing*, $\text{sw}(m) = \sum_d (\delta_{m,d} - \delta_{m,D})$: positive (*swing-up*, a deep discovery) when a move looks worse shallow than it truly is, negative (*swing-down*, a trap) when shallow search flatters it. A position’s total swing, $\sum_m |\text{sw}(m)|$, measures how much the picture changes with depth—how much deliberation the position can reward. The depth at which the played move’s flaw or worth becomes visible is the signal everything below is built on.

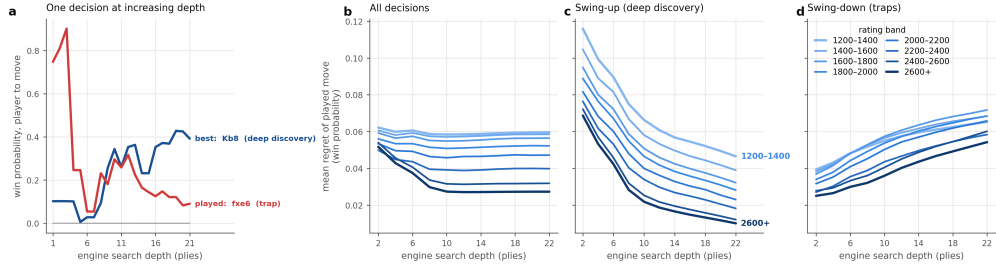


Fig. 1 Skill separates only at depth, and mostly on deep-discovery moves. **a**, A worked example: candidate-move win probabilities across engine search depth for one real decision (grey, other candidates; moves in standard algebraic notation). The played pawn capture fxe6 looks nearly winning at shallow depth and collapses once the refutation surfaces (a *trap*, swing-down); the quiet king move Kb8 reveals its worth only deep (a *deep discovery*, swing-up). **b–d**, Mean regret of the played move across engine search depth, by rating band (light to dark blue: 1200–1400 up to 2600+), within the slow controls (classical and rapid). **b**, All decisions: rating bands are nearly indistinguishable at shallow depth and fan apart as the engine looks deeper. **c**, Swing-up (deep-discovery) decisions: a clean, well-ordered rating fan—the rating signal concentrates here. **d**, Swing-down (trap) decisions: bands bunch and overlap, separating roughly half as much. The rating axis spans online play (lower bands) to elite over-the-board play (upper bands).

Skill lives in the errors, not the successes.

Our primary sample comprises 922,412 decisions by 16,328 players rated 1200 to 2600+, drawn from online play and from elite over-the-board tournaments (Methods). At shallow engine depth, the average error of the played move barely distinguishes a club player from a grandmaster; the gap opens only as the engine looks deeper (Fig. 1b). Because online ratings are pool-specific—a separate rating per time control—we read every skill comparison *within* a control: at full depth, mean regret of the played move falls with rating in classical (0.059 at the weakest band to 0.028 at the strongest), in rapid, and in blitz, but stays nearly flat in bullet (0.070 to 0.055), where no one has time to search deeply. The separating information, moreover, is not spread evenly across mistakes. It is concentrated in *deep-discovery* decisions—swing-up moves whose worth only emerges at depth—which fan out cleanly by rating (Fig. 1c), while trap-like swing-down decisions bunch the rating bands together (Fig. 1d). A single swing-up decision carries roughly twice the rating information of a swing-down one (pool-normalised single-decision $R^2 \approx 0.006$ versus 0.003), an asymmetry that recurs in every replication month below. What matters is not how large an error is, but at what depth the move’s true worth appears.

Depth is not the size of the error.

That claim can be tested directly, without the model. For every erroneous move (full-depth regret ≥ 0.05) define its *exposure depth* as the first search depth at which its regret reaches half its final value—the point at which the flaw comes into view (Methods). Exposure depth is essentially uncorrelated with how large the error eventually proves to be (Spearman $\rho = +0.02$ in the primary month, and between +0.01 and +0.03 in every replication month): how far one must look to see a mistake and how much it costs are separate dimensions. And it is exposure depth that carries the skill signal. Holding error magnitude fixed, stronger players’ errors are exposed deeper.

Among the largest blunders (regret > 0.20), exposure depth rises by 0.93 plies per 1,000 rating points (player-clustered SE 0.07), from about 3.8 plies below 1600 to 4.8 above 2000, and the slope replicates at 1.07, 0.85 and 0.78 plies in the three replication months (all $z > 5$). The effect is graded by severity—weak for small inaccuracies, strongest for large blunders—so the errors that distinguish a strong player from a weak one are not smaller but *better hidden*: mistakes a shallow search would not catch. Skill is legible in where an error surfaces, not in its size.

Depth of satisficing rises with skill—in the pooled likelihood, not the per-decision posterior.

Fitting the latent-depth model (Methods) and reading off each held-out decision's posterior expected depth, the construct appears to behave exactly as a measure of deliberation should, provided one looks within a time control: posterior depth rises with rating in classical (Spearman $\rho = +0.40$), rapid (+0.66) and blitz (+0.45), and is flat in bullet (−0.04). That statistic, however, is the model's own *learned prior* being read back. The prior $\pi_d(c)$ is conditioned on rating, and the correlation of rating with the prior's mean depth (+0.40, +0.67, +0.46, −0.04) matches the posterior's to two decimals, while the correlation of rating with the *update* the played move contributes—posterior minus prior—is +0.01 in every control. A single move says very little about depth. Whether the learned prior is itself supported by the data is a held-out likelihood question, and the answer is: only in slow play. Allowing the depth prior to depend on rating improves held-out move prediction in classical (+0.0020 nats; player-clustered 95% CI [+0.0018, +0.0022]), does nothing in rapid (−0.0000 [−0.0004, +0.0004]), and *worsens* it in blitz (−0.0008 [−0.0011, −0.0005]) and bullet (−0.0023 [−0.0026, −0.0021]). The blitz relationship reported in earlier drafts and in our registrations is therefore an artefact of the prior, and we withdraw it.

The non-circular estimate pools the weak per-decision evidence across each player's decisions instead of leaning on a prior. With the pattern prior evaluated at one fixed rating for every position, no rating in the depth prior, and a flat prior over depth, the depth that best explains a player's moves (Methods) still rises with rating in classical ($\rho = +0.14$, 95% player-bootstrap CI [+0.09, +0.19], 1,477 players) and rapid (+0.20 [+0.15, +0.24]; 2,004), weakly in blitz (+0.12 [+0.08, +0.16]; 2,289), and not at all in bullet (−0.03 [−0.08, +0.01]; 2,071): the registered pattern, recovered with no rating information anywhere in the model (Fig. 2a). Across bands the pooled estimate climbs from about 11 plies at 1200 to 13–17 plies at 2400 and above, and the same signs appear on held-out players alone. Run unchanged on each replication month, the estimator reproduces the pattern every time: classical +0.23 [+0.13, +0.33], +0.31 [+0.21, +0.41] and +0.19 [+0.08, +0.29] in 2026-04, 2026-05 and the prospective 2026-06; rapid +0.20 [+0.11, +0.29], +0.22 [+0.12, +0.31] and +0.28 [+0.19, +0.37]; bullet −0.11, +0.01 and +0.00, never reliably positive; blitz positive with an interval excluding zero in only two months of four (Fig. 2b). Engine plies are an ordinal ruler for effective depth, not literal look-ahead: a Stockfish search to depth 12 is heavily pruned and extended, and think-aloud studies put average human search at only a few plies [18, 25]. What the scale supports is comparison—who satisfices deeper than whom, and when—and that is how it is used throughout. Bullet remains the built-in negative control: when

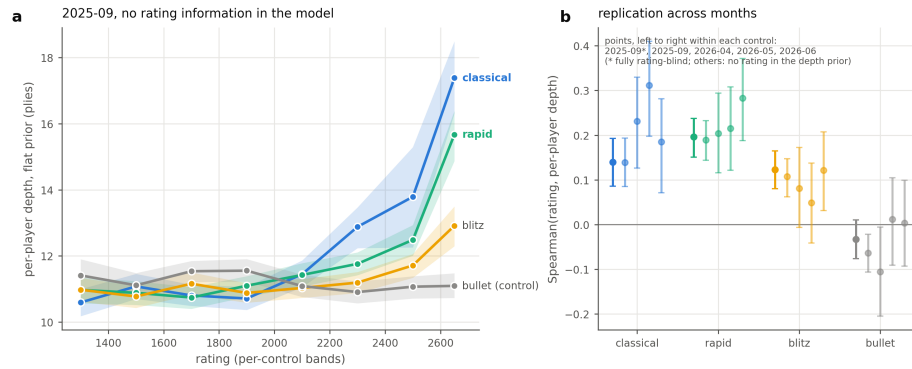


Fig. 2 Depth of satisficing rises with skill under a fully rating-blind estimator. Per-player depth estimated from the move likelihood alone—pattern prior at one fixed rating for every position, no rating in the depth prior, flat prior over depth—pooled over each player’s decisions, by rating band and time control (player-bootstrap 95% CIs). The relationship holds in classical and rapid, weakly in blitz, and is absent in bullet. **a**, 2025-09: per-band mean per-player depth by control. **b**, Spearman correlation of rating with per-player depth, by control, in the primary month (fully rating-blind, starred) and in each replication month run with the same estimator (no rating in the depth prior, flat prior over depth; 95% player-bootstrap CIs). Classical and rapid are positive in every month, bullet is null, and blitz is weak and inconsistent.

nobody can search, the pooled estimate stops separating anybody, which rules out the construct being a re-description of rating.

What the clock buys.

The four controls form a graded ladder of thinking time rather than arbitrary labels. In the online sample the median time a player actually spends on a move falls from about 11 seconds in classical play to 5 in rapid, 3 in blitz, and 1 in bullet—close to an order of magnitude, tracking base budgets from thirty minutes down to one. This ladder is what the depth of satisficing responds to: with more clock a player can carry a line further before committing, so depth rises with skill in classical, rapid, and blitz, where there is time to search, and goes flat in bullet, where a move must be produced in about a second—from recognition rather than search. Within a player, however, the move evidence does not show deeper effective search on slower moves (below); what more time demonstrably buys is *quality*, and it buys more of it for stronger players—the rating gradient of move quality roughly doubles on deliberated moves (Supplementary analysis). The clock sets the ceiling on attainable depth, and skill is expressed in how fully a player uses the room the clock allows—which is why time pressure compresses the skill gap and slow play opens it.

What the move evidence does not show: deeper search on slower moves.

We had taken a clock-free refit as the construct’s convergent validation: with the remaining-clock feature zeroed, held-out posterior depth correlates with observed time-on-move ($\rho = +0.36$, peaking in the middlegame). The decomposition above shows this is not the validation it appears to be. The clock-free prior still carries time class, move number, rating, and position swing, all of which co-vary with time spent; the

prior’s mean depth alone correlates with time at +0.36, identical to the posterior, while the played move’s update correlates at +0.02. Within a time control the posterior is flat or negative except in classical (+0.19, again matching its prior). Two tests that use no depth prior then find no deeper effective search on slower moves. Pooling each player’s decisions and splitting at the player’s own median time, the flat-prior depth of the slower half is not greater than that of the faster half (−0.14 plies; player-bootstrap 95% CI [−0.21, −0.07]; 3,684 player-control cells). And model-free, within player and holding error size, position swing, and move number fixed, errors on slower moves are exposed *shallower*, not deeper (−0.15 plies per *e*-fold of time, player-clustered $z = -6.3$; paired slow minus fast −0.26 plies in classical, −0.19 in rapid). We therefore withdraw the claim that inferred depth tracks thinking time. What deliberation demonstrably buys is quality: the rating gradient of move quality roughly doubles on deliberated moves (Supplementary analysis). Whether that reflects deeper search or better use of the same depth is not something the move evidence, at this density, can decide.

A depth-aware model predicts human moves better, exactly where depth should matter.

If the depth of satisficing is real, knowing it should help predict what a human will play, and the help should be confined to situations where depth can vary and can matter. Both hold. A fusion of a state-only human-move model (Maia-3 [36]) with a mixture-over-depths search model improves held-out cross-entropy over the state-only model by a modest 0.012 nats overall (95% CI [0.008, 0.016], player-clustered bootstrap; top-1 accuracy +0.7 points). The scientific content is not the average but the pre-registered differential: the gain concentrates in classical, high-swing positions (+0.053 nats) and is absent or negative in fast or low-swing play (Fig. 3). The swing-magnitude $\times$ time-control interaction is +0.060 nats (95% CI [0.056, 0.064]), confirmed by a mixed-effects model with players as random effects ($\beta = 0.058$, $p < 10^{-277}$). No state-only model can produce this pattern, however strong [37]; within classical play the gain falls almost entirely on swing-up moves (+0.048 nats)—precisely the choices whose worth a pattern model cannot see because it does not look ahead.

Positions are test items; swing is their discrimination.

These findings admit a measurement-theoretic reading that unifies them. Treat each position as a test item whose *difficulty* is its critical depth—the deepest ply at which a misleading move still looks best—and whose graded *response* is the severity of the move played (best, inaccuracy, mistake, blunder). An explanatory graded-response model [38], with item properties entered as engine-derived covariates in the manner of a linear logistic test model [39] (each natural position occurs only once, so per-item parameters are unidentifiable), recovers the full expected structure: error severity falls with ability ($\beta = -0.13$, $p < 10^{-80}$), rises with item difficulty ($\beta = +0.06$, $p < 10^{-24}$), and—the decisive term—ability discriminates more sharply in high-swing positions (ability $\times$ swing $\beta = -0.08$, $p < 10^{-38}$; Fig. 4a). In item-response language, *swing is item discrimination*: test information concentrates in high-swing decisions, which is exactly why skill is legible in so small a fraction of moves. The graded operating curves

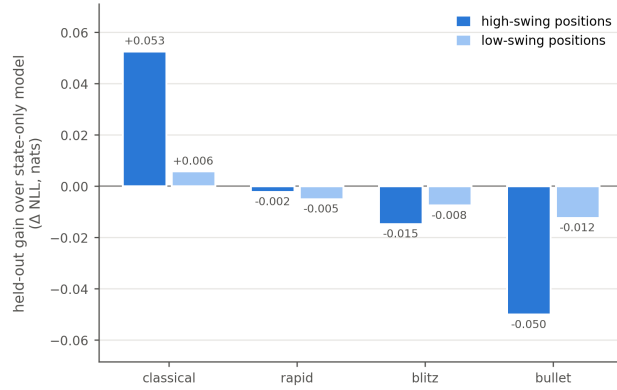


Fig. 3 Depth improves move prediction only where depth can matter. Held-out gain of the depth-aware fusion over the state-only model (nats), by time control and position swing. The gain concentrates in classical, high-swing positions and is absent or negative in fast or low-swing play—the pre-registered differential.

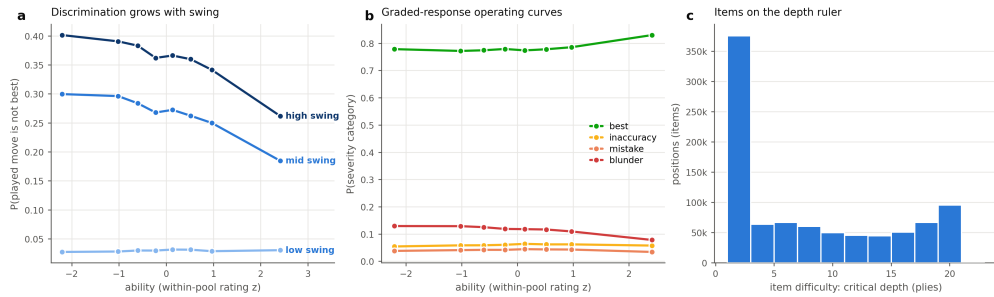


Fig. 4 An item-response reading: swing is item discrimination. **a**, Probability that the played move is not the engine-best, versus ability (within-pool rating z), by position-swing tercile: ability discriminates far more steeply in high-swing positions. **b**, Graded-response operating curves: probability of each severity category versus ability; $P(\text{best})$ rises and $P(\text{blunder})$ falls monotonically. **c**, Items placed on the ply ruler by their critical depth—the same scale on which the depth of satisfying measures person ability.

shift cleanly along the ability axis (Fig. 4b), and item difficulty lives on the same ply ruler as the person-side depth of satisfying (Fig. 4c)—a natural generalisation of rating systems, which are themselves one-parameter item-response models on game outcomes [40–42]. The structure is not a platform artefact. On an independent pool of 99,358 chess.com decisions by 945 players—a different site, a different rating system—the same model reproduces every sign (ability $\beta = -0.08$, difficulty $+0.11$, swing $+0.87$, ability $\times$ swing -0.06 ; all $p < 10^{-13}$), and the same holds on a second Lichess month (2026-04: ability $\times$ swing $\beta = -0.09$, $p < 10^{-40}$)—three independent pools in all. At the present decision density per player, ability is reliable only in aggregate; person-dense, pool-invariant ability estimation is future work.

Engine-derived item difficulty predicts human difficulty.

The item-response reading above is internal: difficulty and ability are both estimated from our own pipeline. An independent criterion exists. The Lichess puzzle database assigns each puzzle a rating on the Glicko scale, estimated from the successes and failures of rated solvers and computed without reference to any engine-depth analysis. We drew a rating-stratified sample of 10,800 well-calibrated puzzles (at least 500 attempts, rating deviation ≤ 80 ; 1,200 per 200-point band from 800 to 2600) from the database snapshot of 2026-09-09, and ran each through the identical engine pipeline (Methods).

Human difficulty tracks the engine’s depth structure. The item’s critical depth—the deepest ply at which the apparent-best move still differs from the full-depth best, the same b_j used above—correlates with the puzzle’s human rating at Spearman $+0.24$, the depth from which the solution becomes and stays best at $+0.26$, and the position’s total swing at $+0.42$ ($n = 10,679$). Puzzle difficulty has two trivial determinants—how many moves the solver must find, and whether the puzzle is a mate—and the relationship survives both: controlling for solution length and for mate and endgame themes, the partial correlations are $+0.17$, $+0.20$ and $+0.26$ (all $p < 10^{-35}$), and adding the depth features to a regression of rating on those controls raises R^2 from 0.293 to 0.330. Mean human rating climbs from 1,598 at critical depth 1 to 1,897 at depth 4 (Fig. 5a); the relationship is clearest between items whose apparent-best move never changes with depth and those where it does, and noisier among the deepest items.

A pattern model supplies the person side and a sharper test. Maia-3’s policy is conditioned on a rating, so running a puzzle through it at a ladder of ratings yields simulated solvers of graded skill, which search not at all. Their solve rate rises monotonically with simulated strength, from 0.60 at 1100 to 0.77 at 2500, and tracks human difficulty across items (Spearman -0.38): much of what makes a puzzle hard is that its solution is hard to see by pattern. But not all of it. Controlling for solution length, theme and the pattern model’s solve rate, engine depth still predicts human difficulty (critical depth $+0.12$, emergence depth $+0.14$, total swing $+0.24$; all $p < 10^{-35}$), and the two kinds of item come apart: puzzles the pattern model solves even at its weakest setting have a mean human rating of 1,574 and a mean critical depth of 2.3, while those it fails at its strongest have a mean rating of 1,988 and a critical depth of 3.2. Deeper items are less findable by pattern at every simulated strength (Fig. 5b; Spearman between critical depth and solve rate -0.20). What a pattern model cannot see, and what makes an item hard for people, is the part that only appears with depth.

Two limits bound this test. Puzzles are announced—the solver is told a decisive move exists, which removes the problem of noticing that a position is critical—so this validates the item side of the measurement, not the person side in natural play. And puzzles are curated towards a unique tactical solution, so they are not a representative sample of positions. What they provide is what the construct otherwise lacks: an item difficulty measured on people, at scale, by someone else.

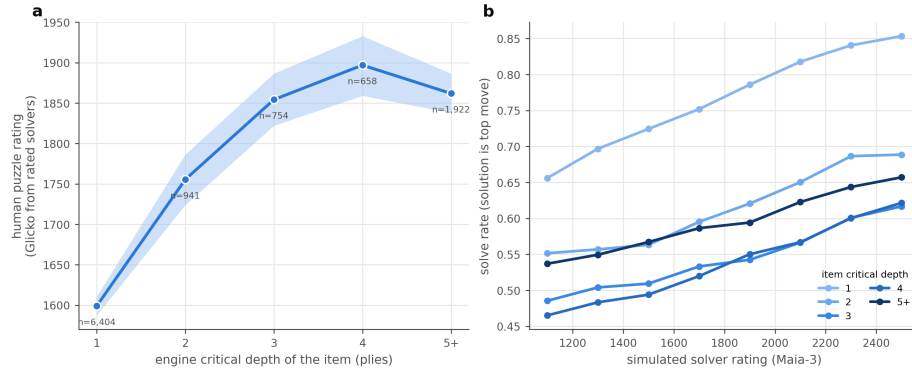


Fig. 5 Engine item difficulty against an independent human criterion. 10,679 rating-stratified Lichess puzzles. **a**, Mean human puzzle rating (Glicko, estimated by Lichess from rated solvers' attempts) by the item's engine critical depth; shaded band, bootstrap 95% CI over puzzles; counts per depth shown. **b**, Person side: solve rate of Maia-3 simulated solvers across a ladder of ratings, by item critical depth. Solve rate rises with simulated skill at every depth, and deeper items sit lower at every strength—what pattern alone cannot find.

Profiles, not a single number, separate expert styles.

Averaged per player from an *elo-free* refit (so that recovering rating is not circular), three behavioural axes—depth of satisficing, trap susceptibility, and deep-discovery rate—summarise a player's style. (An earlier draft carried a fourth, time-elasticity axis, the within-player slope of posterior depth on think-time; it is computed from the posterior, inherits its prior, and on its own recovered $R^2 = 0.01$, so it is dropped.) Across 873 players with at least 100 decisions, the three-axis profile recovers within-pool rating at $R^2 = 0.09$ under nested cross-validation while the depth axis alone recovers none ($R^2 \approx 0$): the two model-free behavioural rates carry the signal, reported on the honest within-pool scale rather than the cross-pool one (which flatters recovery by conflating skill with pool membership). Among elite players the profile also resolves a known embarrassment of one-dimensional scores: a bare depth ranking inverts the true order of the world's best (9 of 15 elite pairs discordant), ranking, for example, Abdusattorov and Nepomniachtchi above Carlsen. The three axes instead attribute each player's strength differently—Nakamura is at once the most trap-resistant and the strongest deep-discoverer of the six, while Abdusattorov and Nepomniachtchi lead on raw depth alone—and the profile's ranking cuts the inversions from 9 to 4 of the 15 pairs (Fig. 6).

The instrument recovers depth it has never seen.

Synthetic agents that satisfice at fixed, known depths, played through the identical pipeline, come back in the right order: recovered depth increases monotonically with planted depth (group-level Spearman = 1.0 across planted depths 4–16), and the estimator independently recovers the agents' decision sharpness ($\hat{\beta} = 7$ against a generating $\beta = 6$; Fig. 7). Recovery is ordinal rather than metric—per-agent estimates are noisy and compressed toward the middle (mean absolute error 3.7 plies on roughly 70 decisions per agent)—but it establishes that the model measures depth rather than

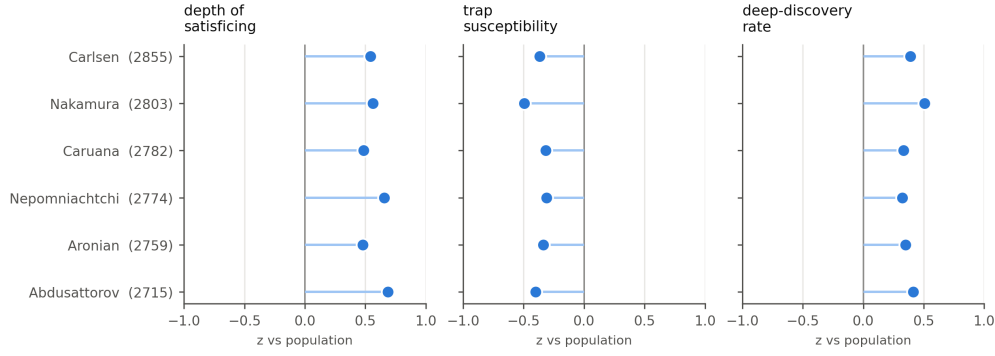


Fig. 6 Three behavioural axes separate elite styles that one number confuses. Profile z-scores (versus the full population) for the six elite players with the most analysed decisions, ordered by rating. A one-dimensional depth score inverts elite ratings (9 of 15 pairs discordant); the three-axis profile recovers within-pool rating across 873 players ($R^2 = 0.09$, nested cross-validation) where depth alone recovers none ($R^2 \approx 0$), and reads the styles directly—for example Nakamura as at once the most trap-resistant and the strongest deep-discoverer of the six.

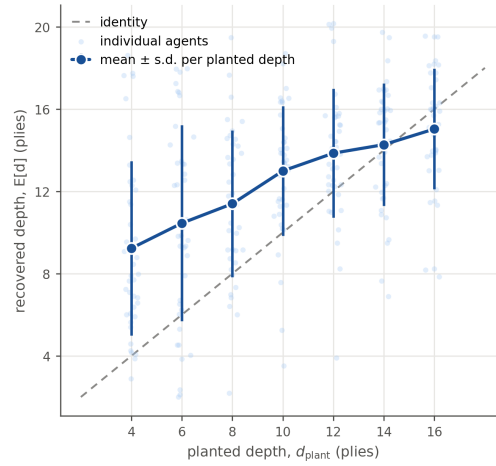


Fig. 7 The instrument recovers planted search depth. Synthetic agents satisficing at fixed depths (4–16 plies), run through the identical pipeline. Pale dots, individual agents; dark points, mean $\pm$ s.d. per planted depth; dashed line, identity. Recovery is monotone (group-level Spearman = 1.0) though ordinal: per-agent estimates are compressed toward the middle (MAE 3.7 plies at ~ 70 decisions per agent).

position difficulty. Recovery requires a flat prior over depth: imposing the human rating-conditioned prior collapses the estimate toward its mean, which is why the two non-circular tests above, and not the posterior’s absolute scale, carry the evidential weight.

A pre-registered programme ending in a prospective confirmation.

We pre-registered hypotheses, pipeline, and decision rules, and then ran the frozen pipeline once on each of three untouched Lichess months [35]. The first replication

(2026-05; 179,360 decisions; OSF [KB4ZQ](#)) reproduced the registered clock-free correlation ($\rho = +0.30$, middlegame peak; a statistic we now know to be carried by the prior, above), and the registered swing $\times$ time-control interaction held ($+0.049$ nats, mixed-effects $p < 10^{-36}$)—but failed its own strict rule because one arm of one primary (blitz, registered as positive) came out flat, and it exposed a swing-label convention error: the single-decision rating signal sits in swing-*up* decisions, not swing-down as first registered. We disclosed both, corrected the hypotheses, and registered a second, fully prospective protocol (OSF [A6NYK](#)) whose primary dataset—Lichess 2026-06—did not yet exist at registration. A robustness month processed under the same registration (2026-04; 168,654 decisions) meanwhile reproduced the registered rating relationship (classical $\rho = +0.47$, rapid $+0.54$), the clock-free correlation ($+0.40$), and the interaction ($+0.076$ nats, $p < 10^{-48}$).

The prospective month then arrived and was processed once. All three primary hypotheses replicated, satisfying the registered decision rule. On 203,248 decisions from 3,707 players: depth rose with rating within every slow control (classical $\rho = +0.23$, 95% CI $[+0.15, +0.35]$; rapid $+0.72$ $[+0.59, +0.79]$) with bullet null as predicted ($+0.08$ $[-0.03, +0.19]$); the registered clock-free correlation with thinking time held ($\rho = +0.32$, middlegame peak $+0.36$; a prior-carried statistic, below); and the depth-aware model beat the state-only model overall ($+0.010$ nats, 95% CI $[+0.003, +0.017]$) with the registered interaction intact ($+0.068$ nats $[+0.055, +0.079]$; mixed-effects $\beta = +0.059$, $p < 10^{-42}$). Two secondary outcomes are worth reporting straight. Blitz, which we had over-corrected to “null” after its flat 2026-05 showing, came back clearly positive ($\rho = +0.39$ $[+0.25, +0.50]$); across the four months the blitz relationship now reads $+0.45$, ≈ 0 , $+0.19$, $+0.39$ —real but unstable, in contrast to the robust slow controls and the reliably null bullet. And the deep-discovery asymmetry replicated in its corrected direction (swing-up $R^2 = 0.0049$ versus swing-down 0.0027), as it has in every month examined.

Two of the three registered primaries must now be read differently. H1 (posterior depth versus rating) and H3 (clock-free posterior depth versus time) were operationalised, in both registrations, as correlations of the model’s per-decision posterior, which the decomposition above shows to be dominated by the learned context prior. They replicated as registered, but as tests of the prior rather than of the move evidence; we registered them in good faith and discovered the problem only in a later audit. H4, the prediction differential, is a likelihood test and stands as registered. Re-tested with the rating-blind pooled estimator, H1 survives: classical and rapid are positive with intervals excluding zero in every month, including the prospective one (classical $+0.19$ $[+0.08, +0.29]$, rapid $+0.28$ $[+0.19, +0.37]$ on 2026-06), and bullet is null in every month (Fig. 2b). H3 does not survive the non-circular within-player tests reported above, and we record it as a withdrawn claim rather than a confirmed one. The registered confirmation was, in the strict sense, predicted in public before the behaviour occurred; the audit shows which of its parts that prediction actually tested.

3 Discussion

Errors are usually the noise of behavioural data. Here they are the signal. Where success is widely shared, the fine structure of failure—the depth at which a flaw becomes visible—turns out to be a stable, measurable signature of how a person reasons: it scales with expertise under an estimator given no rating information, is absent when the clock removes the possibility of search, and improves prediction exactly in the regime where deep reasoning has room to operate. The pattern supports a specific mechanism: people commit to the best option found at the search horizon they reach, rather than integrating over the whole trajectory—consistent with theoretical results that level- k predictions are independent of higher-order belief tails [16], and connecting satisficing to model-based control [43], decision-tree pruning [44], and bounded evidence accumulation [45]. Where planning depth has been measured before, it required controlled laboratory games and explicit search models [7, 17]; here the same construct is read non-invasively from natural, high-stakes play at population scale, against a depth-resolved value standard, and its registered prediction differential survives a prospective test. The item-response formulation adds a measurement-theoretic dividend: positions become items with engine-derived difficulty and discrimination, so people and problems share one ply-denominated scale—rating systems generalised from game outcomes to individual decisions.

The result also speaks to the oldest question in the psychology of chess: whether experts search deeper or merely search better [18, 25, 28, 29]. The depth of satisficing measures *effective* depth—how far the chosen move survives scrutiny—and is deliberately agnostic about whether that depth is reached by looking further down a line or by selecting, from pattern, lines that do not fail early. The data suggest both operate and can be told apart. Skill is visible even on moves played in a second, where only pattern can act; yet expert errors of a given cost are better hidden rather than smaller, and, in a supplementary analysis, the skill gradient in move quality roughly doubles when players deliberate rather than move at once. De Groot’s null and Charness’s positive finding are then reconciled by the clock: with little time, strong and weak players look alike in depth and differ in pattern; given time, the strong convert it into depth and the weak cannot.

The individual profiles suggest what the construct is for. Expertise is not a scalar: two players of equal rating can owe their strength to depth, to trap resistance, to deep discovery—echoing evidence that expert flexibility itself varies [46]. An agent that estimates a person’s depth of satisficing could, in principle, select problems pitched just beyond it; a rating system built on decisions rather than outcomes could equate skill across pools that never play each other.

The limitations are the flip side of the design. Engine search depth is a *validated proxy* for human selective search—validated by recovery of planted depth and by the model-free exposure-depth and prediction results, not, as we first thought, by convergence with thinking time—but it is not isomorphic to human search, which is narrow, uneven, and pattern-guided. The per-decision posterior is dominated by its context prior: the played move shifts it by little, so correlations of the posterior with anything in that context—rating, time class, remaining clock—are largely the prior read back, and two of our registered statistics were of that kind. Claims about depth

therefore rest on estimators that pool evidence across a player’s decisions with no prior over depth, on model-free exposure depth, on the prediction differential, and on planted-depth recovery, never on per-decision posteriors; a time-elasticity profile axis used in an earlier draft inherited the same limitation and has been dropped. The average predictive gain is small, and the case rests on its registered differential rather than its size. The blitz rating–depth relationship is not supported once the prior is removed and should not be quoted. And chess players are a highly trained, self-selected population in an adversarial task; the approach extends in principle to any domain with skill-graded decisions and a depth-resolved value standard—a value network queried at increasing depth in Go, or staged value estimates in planning tasks—but that generality remains to be demonstrated.

4 Methods

Data.

The primary dataset comprises 922,412 decisions by 16,328 players: online games from the Lichess open database (month 2025-09), balanced-sampled across rating bands (200-point bins, 1200–2600+) and time classes (bullet, blitz, rapid, classical), with per-move clock times retained. Time classes follow Lichess’s convention, by estimated game duration (initial time plus $40 \times$ the increment): bullet under 3 minutes, blitz 3–8, rapid 8–25, and classical 25 minutes or more. Each class therefore pools nearby formats—1+0 and 1+1 bullet, for instance, both fall in the bullet class despite an appreciable difference in effective time—so residual within-class heterogeneity remains, and every skill comparison in this paper is made within class, never across; plus an elite over-the-board stratum of 371,279 decisions from 1,261 titled players in broadcast tournament games, which supports the individual-profile case study. Replication datasets—Lichess 2026-05 (179,360 decisions), 2026-04 (168,654), and the prospective 2026-06 (203,248 decisions, 3,707 players, 113,421 classical)—were each sampled by the same frozen pipeline (identical seed, cell targets, and engine settings) and processed once. The independent-pool item-response check uses 99,358 decisions by 945 titled chess.com players harvested via the public Published-Data API. Synthetic agents (below) provide ground truth. All decisions are restricted to plies 9–120 to skip opening-book play; no further selection is applied, so every non-book decision in a sampled game is analysed and the sample is not conditioned on position difficulty or on the quality of the move played.

Engine analysis and value scaling.

Positions are analysed with Stockfish 18 in multi-PV mode, recording each candidate move m_i ’s evaluation at every depth $d = 1 \dots D$ ($D=21$, $K=9$). The candidate set is the union of the top- K across all depths, so trap moves keep a full trajectory. Because consecutive plies of each game are analysed, the value of a played move falling outside the top- K is read off from the position it leads to (the next analysed decision): its win probability at depth d equals one minus the successor’s best win probability at depth $d-1$, matching the other candidates’ negamax depth. Where no analysed successor exists (game end, final sampled ply) the played move takes the

weakest retained candidate’s win probability at each depth. With UCI_ShowWDL the engine reports win/draw/loss counts (w, dr, ℓ) per mille, giving the win probability $P_{i,d} = (w + \frac{1}{2} dr)/1000$ directly—no logistic or centipawn scaling. Depth- d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \geq 0. \quad (1)$$

Being a difference against the best move *at each depth*, δ is invariant to anything that shifts all moves together (position difficulty, evaluation scale), isolating the choice problem. Searches run to depth D or a 2×10^8 -node cap, whichever comes first; the 0.09% of positions that stop short have their unreached depths masked in the likelihood (reached depth recorded) rather than imputed.

Pattern prior.

The human-pattern prior $q_i(p, c)$ is the Maia-3 policy [36] at the side-to-move’s rating, read from the policy head as a full distribution over legal moves.

Latent-depth choice model.

A player who has effectively searched to depth d chooses by a soft-max over depth- d regret fused with the pattern prior as a product of experts,

$$P(\text{move}=i \mid d, p, c) = \frac{\exp(-\beta \delta_{i,d} + \alpha \log q_i)}{\sum_j \exp(-\beta \delta_{j,d} + \alpha \log q_j)}. \quad (2)$$

Effective depth is latent with distribution $\pi_d(p, c)$, emitted by a small network over a coarse depth grid conditioned on context c (rating, time class, clock, move number, total swing). The move likelihood marginalises depth, $P(\text{move}=i \mid p, c) = \sum_d \pi_d(p, c) P(\text{move}=i \mid d, p, c)$, and parameters minimise the negative log-likelihood of the played move with an entropy penalty on π for identifiability and β tied globally. Data are split by player throughout.

Depth of satisfying.

The posterior over effective depth given the observed move,

$$r_d = P(d \mid y, p, c) = \frac{\pi_d P(y \mid d)}{\sum_{d'} \pi_{d'} P(y \mid d')}, \quad \hat{d} = \sum_d d r_d, \quad (3)$$

gives a per-decision depth with uncertainty. Lichess ratings are pool-specific (a separate Glicko-2 rating per time control, not comparable across controls), so no analysis pools ratings across controls; doing so would also confound skill with available clock. Per-band estimates carry player-clustered bootstrap 95% intervals (1,000 resamples of players) [47].

Exposure depth.

For the model-free discriminant test, an erroneous move (full-depth regret ≥ 0.05) has exposure depth equal to the first grid depth at which its regret reaches half of

its full-depth value. Within bins of full-depth regret (0.05–0.10, 0.10–0.20, >0.20), exposure depth is regressed on rating (per 1,000 points) and move number with player-clustered standard errors, on online classical and rapid decisions; the same script is run unchanged on each replication month.

Relation to the curve-intersection estimator.

The construct originates in a population-level heuristic [33]: average the played move’s error and the engine move’s error against depth, and read the depth of satisficing off the curves’ crossing. That estimator returns one depth per cohort, carries no per-decision uncertainty, presupposes a single time control, and inherits the fit (cohort depths move by roughly a ply under reasonable fitting choices). We keep the underlying signal—a played move whose error grows with depth is exactly the move *swing* of Biswas and Regan [33]—and replace the crossing with the generative posterior r_d : a per-decision quantity with credible intervals that supports within-control and within-player contrasts and is identifiable against synthetic ground truth. The population curve is stable only because it averages thousands of weak per-move signals, which the crossing view conceals and our identifiability test makes explicit: recovered depth is ordinal and prior-compressed, which is why the rating-blind pooled estimator, the model-free tests, and the identifiability test—not the posterior’s absolute scale—carry the claims. Where the two views can be compared they agree—stronger players satisfice deeper and err less—now within time control, at far larger scale, on win-probability evaluation, with a human-pattern prior, and under a registered replication programme.

Item-response formulation.

Each decision is an item. Difficulty b_j is the deepest grid depth at which the apparent-best move differs from the full-depth best move—the critical depth d_i of Biswas and Regan [33], in plies; the ordered response is the played move’s full-depth regret binned as best (≤ 0.02), inaccuracy (≤ 0.05), mistake (≤ 0.10), or blunder (> 0.10) in win-probability units. We fit an explanatory proportional-odds graded-response model [38, 39] with ability (within-pool rating z), b_j , position swing, and ability $\times$ swing as predictors; item properties enter as covariates rather than free parameters because most natural positions occur exactly once. The MLE uses a random subsample; response curves are computed on all decisions. Per-player ability has low split-half reliability at this decision density, so effects are reported at the item and rating-band level.

Puzzle validation.

Puzzles come from the Lichess puzzle database (snapshot 2026-09-09; 6,100,952 puzzles), restricted to well-calibrated items (at least 500 recorded attempts, rating deviation ≤ 80) and sampled at 1,200 per 200-point rating band from 800 to 2600. A puzzle’s position is the one shown to the solver—after the opponent’s setup move—and its solution is the solver’s first move; both are analysed by the identical engine pipeline, so item difficulty is computed exactly as in the item-response analysis. Simulated solvers are the Maia-3 policy evaluated at eight ratings from 1100 to 2500 on the same positions, scored by whether the solution is the model’s top move. Regressions of

puzzle rating use heteroskedasticity-robust standard errors; puzzles are independent items, so no clustering is applied.	737
	738
	739
<i>Profiles and rating recovery.</i>	740
Per player with at least 100 decisions: mean $\hat{d}$; trap susceptibility (rate of played swing-down moves, weighted by shallow attractiveness); deep-discovery rate (rate of played swing-up moves). The depth axis comes from a model fit with the rating feature removed, keeping recovery non-circular; a time-elasticity axis (within-player slope of $\hat{d}$ on log think-time) was dropped because it is a function of the posterior's clock-dependent prior. Rating is predicted from the standardised profile by ridge regression under nested five-fold cross-validation (inner folds select the penalty), within pool (z-scored per time control, over-the-board play its own pool). The elite case study reports the standardised axes for the six broadcast players with the most analysed decisions.	741
	742
	743
	744
	745
	746
	747
	748
	749
	750
<i>Model comparison and the registered differential.</i>	751
State-only ($\beta=0$), search-only ($\alpha=0$), and fusion predictors are fit under one player split and compared on held-out cross-entropy and top-1 match. The registered hypothesis is an interaction: the fusion's gain over state-only is larger in high-swing than low-swing positions and in classical than blitz. It is tested with a player-clustered bootstrap and a linear mixed-effects model with player random intercepts.	752
	753
	754
	755
	756
	757
<i>Decomposition, rating-blind estimation, and recovery validation.</i>	758
Decomposition: for a fitted model, each held-out decision's prior mean depth $\sum_d d \pi_d(c)$ is compared with its posterior $\hat{d}$; the update is their difference. Held-out likelihood tests refit the model with the rating (or remaining-clock) context feature zeroed and compare per-decision negative log-likelihood, with player-clustered bootstrap intervals. Rating-blind per-player depth: the pattern prior is recomputed with Maia-3 conditioned on one fixed rating (1,950, the online sample median) for every position; the model is refit with the rating context feature zeroed to obtain (α, β) ; for each (player, control) with at least 30 decisions the depth log-likelihoods $\log P(y_i d)$ are summed over decisions and a flat prior over the grid gives $\hat{d}$; Spearman against rating carries a 2,000-resample player bootstrap. Replication months use their existing rating-conditioned pattern prior with the same rating-free depth estimation; on the primary month the two versions agree (classical +0.139 versus +0.140). The same estimator, split at each player's median time-on-move, gives the within-player slow-versus-fast test; its model-free counterpart regresses exposure depth on log time-on-move with player fixed effects and player-clustered standard errors, controlling for error magnitude, position swing, and move number. Recovery: agents play a soft-max ($\beta_{\text{gen}}=6$) over fixed-depth win probabilities; their games pass through the identical pipeline, and a per-agent flat-prior posterior over depth (search likelihood, with the population β fit) is compared with the planted depth.	759
	760
	761
	762
	763
	764
	765
	766
	767
	768
	769
	770
	771
	772
	773
	774
	775
	776
	777
	778
	779
<i>Pre-registered replication programme.</i>	780
Hypotheses, pipeline, sampling seed, and the decision rule (success iff all three primary hypotheses replicate directionally with stated intervals) were registered before	781
	782

783 analysing each replication month (first registration: OSF KB4ZQ; corrected and
784 prospective second registration: OSF A6NYK, whose primary month 2026-06 was reg-
785 istered before Lichess published the data). Each month was processed exactly once
786 by the frozen pipeline. Deviations are disclosed in Results: the first registration’s blitz
787 prediction was too strong and its swing-direction convention reversed; the second regis-
788 tration’s blitz-null prediction proved too weak in turn (blitz positive in the prospective
789 month). Neither deviation touches the primary hypotheses.

790

791 ***Reproducibility.***

792 Stockfish 18 (shared NNUE), Threads 1 per process across many single-thread engines,
793 Hash 256 MB, $D=21$, $K=9$, 2×10^8 -node cap, driven through the `python-chess`
794 UCI bridge (v1.11), which also provides all board and move representation; Maia-3
795 (maia3-79m) [36]; depth grid $\{2, 4, \dots, 22\}$; sampling seed 17 and 60 games per (rating
796 band $\times$ time class) cell in every month; train/test split by player; model fits seeded.
797 All analyses ran on a single dual-socket 64-core server with one GPU; the engine stage
798 dominates the compute at roughly 1,400 core-hours per month of data, and all model
799 fitting and analysis complete within about an hour. Strata and the primary interaction
800 were fixed before held-out evaluation; post-hoc choices are labelled exploratory.

801

802

803 ***Ethics.***

804 This study is a secondary analysis of publicly available, pseudonymous game records;
805 no participants were recruited, no interaction or intervention took place, and no private
806 or identifiable personal information was collected or processed. Online players appear
807 only under their self-chosen public usernames; the named elite players analysed in
808 the individual profiles are public figures whose over-the-board tournament games are
809 published as a matter of course. As research limited to secondary analysis of public
810 data, the study did not require prior ethics approval and no formal ethics review was
811 sought; informed consent was not required for the same reason.

812

813 ***Data availability.***

814 All game records derive from public sources: the Lichess open database (<https://database.lichess.org>;
815 months 2025-09, 2026-04, 2026-05, 2026-06), openly licensed
816 Lichess broadcast exports of over-the-board play, and public games retrieved via
817 the chess.com Published-Data API (<https://api.chess.com>). The derived analysis
818 datasets—per-position engine depth trajectories, Maia-3 policies, and the model ten-
819 sors behind every result, for the primary month and all three replication months—are
820 deposited in the study’s Open Science Framework project (<https://osf.io/ruhy8>) under
821 a CC-BY licence; the deposit DOI will be minted on publication.

822

823

824 ***Code availability.***

825 The full pipeline (engine driver, dataset builder, model, confirmatory replication
826 script, and all figure code) and configuration files are openly available under the MIT
827 licence at <https://github.com/tamalrkm/depth-of-satisficing>. The pre-registrations
828 are archived in the same OSF project (registrations [10.17605/OSF.IO/KB4ZQ](https://osf.io/KB4ZQ) and

[10.17605/OSF.IO/A6NYK](https://doi.org/10.17605/OSF.IO/A6NYK)). Stockfish 18 and Maia-3 are third-party open-source software.

Acknowledgements.

The authors gratefully acknowledge the support from the Department of Science and Technology (DST), Government of India, through the DST-FIST grant [Sanction No.: SR/FST/MS-I/2022/116].

Author contributions.

T.M. conceived the study, designed and implemented the latent-depth model and the full analysis pipeline, ran the experiments, pre-registered and ran the replication programme, and drafted the manuscript. K.W.R. contributed the intrinsic-performance-rating and move-swing framework underlying the construct, advised on chess-cognition methodology and validation design, and critically reviewed the manuscript. Both authors approved the final version.

Competing interests.

The authors declare no competing interests.

References

- [1] Simon, H. A. A behavioral model of rational choice. *The Quarterly Journal of Economics* **69**, 99–118 (1955).
- [2] Simon, H. A. Rational choice and the structure of the environment. *Psychological Review* **63**, 129–138 (1956).
- [3] Gigerenzer, G. & Goldstein, D. G. Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review* **103**, 650–669 (1996).
- [4] Russell, S. & Wefald, E. Principles of metareasoning. *Artificial Intelligence* **49**, 361–395 (1991).
- [5] Griffiths, T. L., Lieder, F. & Goodman, N. D. Rational use of cognitive resources: Levels of analysis between the computational and the algorithmic. *Topics in Cognitive Science* **7**, 217–229 (2015).
- [6] Lieder, F. & Griffiths, T. L. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences* **43**, e1 (2020).
- [7] Callaway, F. *et al.* Rational use of cognitive resources in human planning. *Nature Human Behaviour* **6**, 1112–1125 (2022).
- [8] Tversky, A. & Kahneman, D. Judgment under uncertainty: Heuristics and biases. *Science* **185**, 1124–1131 (1974).

- 875 [9] Kahneman, D. & Tversky, A. Prospect theory: An analysis of decision under risk.
876 *Econometrica* **47**, 263–291 (1979).
877
- 878 [10] Stahl, D. O. & Wilson, P. W. On players’ models of other players: Theory and
879 experimental evidence. *Games and Economic Behavior* **10**, 218–254 (1995).
880
- 881 [11] Nagel, R. Unraveling in guessing games: An experimental study. *The American*
882 *Economic Review* **85**, 1313–1326 (1995).
883
- 884 [12] Costa-Gomes, M., Crawford, V. P. & Broseta, B. Cognition and behavior in
885 normal-form games: An experimental study. *Econometrica* **69**, 1193–1235 (2001).
886
- 887 [13] Camerer, C. F., Ho, T.-H. & Chong, J.-K. A cognitive hierarchy model of games.
888 *The Quarterly Journal of Economics* **119**, 861–898 (2004).
889
- 890 [14] Costa-Gomes, M. A. & Crawford, V. P. Cognition and behavior in two-person
891 guessing games: An experimental study. *The American Economic Review* **96**,
892 1737–1768 (2006).
893
- 894 [15] Crawford, V. P., Costa-Gomes, M. A. & Iriberri, N. Structural models of
895 nonequilibrium strategic thinking: Theory, evidence, and applications. *Journal*
896 *of Economic Literature* **51**, 5–62 (2013).
897
- 898 [16] Strzalecki, T. Depth of reasoning and higher order beliefs. *Journal of Economic*
899 *Behavior & Organization* **108**, 108–122 (2014).
900
- 901 [17] van Opheusden, B. *et al.* Expertise increases planning depth in human gameplay.
902 *Nature* **618**, 1000–1005 (2023).
903
- 904 [18] de Groot, A. D. *Thought and Choice in Chess* (Mouton, The Hague, 1965).
905
- 906 [19] Newell, A. & Simon, H. A. *Human Problem Solving* (Prentice-Hall, Englewood
907 Cliffs, NJ, 1972).
908
- 909 [20] Chase, W. G. & Simon, H. A. Perception in chess. *Cognitive Psychology* **4**, 55–81
910 (1973).
911
- 912 [21] Gobet, F. & Simon, H. A. Templates in chess memory: A mechanism for recalling
913 several boards. *Cognitive Psychology* **31**, 1–40 (1996).
914
- 915 [22] Ericsson, K. A., Krampe, R. T. & Tesch-Römer, C. The role of deliberate practice
916 in the acquisition of expert performance. *Psychological Review* **100**, 363–406
917 (1993).
918
- 919 [23] Campbell, M., Hoane Jr., A. J. & Hsu, F.-h. Deep Blue. *Artificial Intelligence*
920 **134**, 57–83 (2002).

- [24] Silver, D. *et al.* A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science* **362**, 1140–1144 (2018). 921–923
- [25] Charness, N. Search in chess: Age and skill differences. *Journal of Experimental Psychology: Human Perception and Performance* **7**, 467–476 (1981). 924–925
- [26] Holding, D. H. *The Psychology of Chess Skill* (Lawrence Erlbaum Associates, Hillsdale, NJ, 1985). 926–927
- [27] Campitelli, G. & Gobet, F. Adaptive expert decision making: Skilled chess players search more and deeper. *ICGA Journal* **27**, 209–216 (2004). 928–931
- [28] Gobet, F. Chess players’ thinking revisited. *Swiss Journal of Psychology* **57**, 18–32 (1998). 932–933
- [29] Connors, M. H., Burns, B. D. & Campitelli, G. Expertise in complex decision making: The role of search in chess 70 years after de Groot. *Cognitive Science* **35**, 1567–1579 (2011). 934–935
- [30] Guid, M. & Bratko, I. Computer analysis of world chess champions. *ICGA Journal* **29**, 65–73 (2006). 936–937
- [31] Regan, K. W. & Haworth, G. M. *Intrinsic chess ratings*. In *Proceedings of the 25th AAAI Conference on Artificial Intelligence*, 834–839 (2011). 938–939
- [32] Anderson, A., Kleinberg, J. & Mullainathan, S. *Assessing human error against a benchmark of perfection*. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 705–714 (2016). 940–941
- [33] Biswas, T. T. & Regan, K. W. *Measuring level-k reasoning, satisficing, and human error in game-play data*. In *2015 IEEE 14th International Conference on Machine Learning and Applications (ICMLA)*, 941–947 (2015). 942–943
- [34] McIlroy-Young, R., Sen, S., Kleinberg, J. & Anderson, A. *Aligning superhuman AI with human behavior: Chess as a model system*. In *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 1677–1687 (2020). 944–945
- [35] Nosek, B. A., Ebersole, C. R., DeHaven, A. C. & Mellor, D. T. The preregistration revolution. *Proceedings of the National Academy of Sciences* **115**, 2600–2606 (2018). 946–947
- [36] Monroe, D., Eilender, G., Chalmers, P., Tang, Z. & Anderson, A. *Chessformer: A unified architecture for chess modeling*. In *International Conference on Learning Representations (ICLR)* (2026). arXiv:2605.19091. 948–949
- [37] Ruoss, A. *et al.* *Amortized planning with large-scale transformers: A case study on chess*. In *Advances in Neural Information Processing Systems (NeurIPS)*, 950–951

967 Vol. 37, 65765–65790 (2024). arXiv:2402.04494.
968
969 [38] Samejima, F. Estimation of latent ability using a response pattern of graded
970 scores. *Psychometrika Monograph Supplement* **34**, 1–97 (1969).
971
972 [39] Fischer, G. H. The linear logistic test model as an instrument in educational
973 research. *Acta Psychologica* **37**, 359–374 (1973).
974
975 [40] Elo, A. E. *The Rating of Chess Players, Past and Present* (Arco, New York,
976 1978).
977
978 [41] Rasch, G. *Probabilistic Models for Some Intelligence and Attainment Tests*
979 (Danish Institute for Educational Research, Copenhagen, 1960).
980
981 [42] Embretson, S. E. & Reise, S. P. *Item Response Theory for Psychologists*
982 (Lawrence Erlbaum Associates, Mahwah, NJ, 2000).
983
984 [43] Daw, N. D., Niv, Y. & Dayan, P. Uncertainty-based competition between
985 prefrontal and dorsolateral striatal systems for behavioral control. *Nature*
986 *Neuroscience* **8**, 1704–1711 (2005).
987
988 [44] Huys, Q. J. M. *et al.* Bonsai trees in your head: How the Pavlovian system sculpts
989 goal-directed choices by pruning decision trees. *PLoS Computational Biology* **8**,
990 e1002410 (2012).
991
992 [45] Ratcliff, R. & McKoon, G. The diffusion decision model: Theory and data for
993 two-choice decision tasks. *Neural Computation* **20**, 873–922 (2008).
994
995 [46] Bilalić, M., McLeod, P. & Gobet, F. Inflexibility of experts—Reality or myth?
996 Quantifying the Einstellung effect in chess masters. *Cognitive Psychology* **56**,
997 73–102 (2008).
998
999 [47] Efron, B. Bootstrap methods: Another look at the jackknife. *The Annals of*
1000 *Statistics* **7**, 1–26 (1979).
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012